

Bonus: Movie Watchlist with `std::vector`

Write a C++ program that manages a small movie watchlist.

Learning goals

- Create and use a `vector<string>`
- Add elements with `push_back`
- Insert one element with `insert`
- Remove one element with `erase`
- Traverse a vector with an iterator
- Use `size`, `capacity`, and `reserve`

Task

Write a program that does the following:

1. Create an empty `vector<string>` called `watchlist`.
2. Reserve space for 10 movies.
3. Add these movies using `push_back`:
 - "Casablanca"
 - "The Matrix"
 - "Spirited Away"
 - "Rear Window"
4. Insert "The Godfather" before "The Matrix".
5. Remove "Rear Window" from the list.
6. Print the final watchlist using an iterator.
7. Print the vector size and capacity.

Expected output

Movie watchlist:

1. Casablanca
2. The Godfather
3. The Matrix
4. Spirited Away

Size: 4

Capacity: 10

Starter code

```
#include <iostream>
#include <vector>
#include <string>
using namespace std;

int main()
{
    // Create empty vector

    // Reserve space for 10 movies

    // Add four movies

    // Insert "The Godfather" before "The Matrix"

    // Remove "Rear Window"

    // Print final watchlist using an iterator

    // Print size and capacity

    return 0;
}
```

Notes for checking

- The program should compile without warnings or errors.

- The final vector should contain exactly 4 movies.
- "The Godfather" should appear before "The Matrix".
- "Rear Window" should not appear in the final output.
- The capacity should be at least 10, and normally exactly 10 after `reserve(10)`.

Submission

Submit the URL to your solution in [OneCompiler.com/cpp](https://onecompiler.com/cpp)